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纤维分布对混杂纤维增强聚合物(HFRP)筋拉伸性能影响的试验和数值模拟研究

Role of fiber distribution on tensile properties of hybrid fiber-reinforced polymer (HFRP) bars through experimental and numerical simulation study

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ABSTRACT

The distribution pattern of fiber significantly influences the tensile performance of HFRP bars. This study designed and produced two types of hybrid FRP bars with distinct carbon fiber configurations: peripherally distributed H_1FRP and centrally concentrated H_2FRP. Acoustic emission technology was applied to investigate the synergetic mechanism of carbon and glass fibers with different distribution patterns. Comparative analysis of signal accumulation during tensile tests confirmed that peripheral carbon fiber distribution effectively suppressed the propagation of fiber/resin interfacial debonding and matrix cracking. H_1FRP bars exhibited 17 % lower tensile strength than H_2FRP bars (983.3 MPa vs 1185.3 MPa), while demonstrating superior ductility and unique yielding characteristics. X-ray computed tomography (CT) quantification revealed that outer fiber layers showed significantly lower volume retention rates (R_{C1} = 81.93 %; R_{G2} = 82.73 %) compared to inner layers (R_{C2} = 93.87 %; R_{G1} = 97.22 %). A shear-lag theory-based finite element model verified transverse shear stress-induced supplementary damage, achieving 93.7 % consistency with experimental results. Visualization-driven simulations provided elastic modulus correction coefficients (α_1 = 0.92, β_1 = 0.357; α_2 = 0.85, β_2 = 0). The constitutive model incorporating fiber distribution parameters demonstrated 15 % higher prediction accuracy than conventional models, establishing new theoretical foundations for performance-oriented design of HFRP bars.

1. Introduction

In reinforced concrete structures, corrosion of steel reinforcement leads to diminished load-bearing capacity and stiffness, with severe cases causing structural collapse and substantial economic losses [1]. Aggravated corrosion rates are observed in marine and island environments due to inherent concrete porosity [2]. Fiber reinforced polymer (FRP) bars, composed of non-metallic fibers and resin matrices, demonstrate superior corrosion resistance in chloride-rich environments, positioning them as promising steel alternatives [3]. Additional advantages including high strength-to-weight ratio, long-term durability, and non-magnetic properties further support their engineering applications [4].

Conventional FRP bars are manufactured through pultrusion processes using single type fibers (e.g., carbon, glass, basalt, or

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aramid) embedded in thermosetting resins. While fibers provide primary load-bearing functionality, the resin matrix ensures fiber bonding and stress transfer [5]. However, their practical application remains constrained by brittle fracture mechanisms under tensile loading [6]. To end this, HFRP bars incorporating fibers with distinct stiffness profiles have been developed to achieve progressive fracture and enhanced plasticity [7]. The inclusion of high-stiffness fibers simultaneously improves overall structural rigidity [8]. Current hybridization strategies primarily involve carbon/glass [9] or carbon/basalt [10–13] combinations. Although carbon fibers exhibit superior mechanical properties, corrosion resistance, and anisotropy compared to glass or basalt fibers, their high cost necessitates partial substitution approaches [14]. This optimization balances performance enhancement (mechanical properties, stiffness, corrosion resistance, and fracture ductility) with economic feasibility [15–18]. Consequently, carbon fiber-hybridized HFRP bars demonstrate superior mechanical performance and broader application prospects compared to Single-fiber-type FRP bars [19].

Numerous researchers have investigated ductility enhancement in FRP bars through fiber hybridization [20]. Critical parameters include volume fraction of low-elongation fibers, spatial distribution patterns, and fiber type combinations [21]. Bakis et al. [22], Ali et al. [23] and Jones et al. [24] systematically examined how high-stiffness fiber content and distribution configurations influence tensile ductility, revealing that reduced volume fractions and dispersed distributions optimize ductile behavior. Zhang et al. [25] developed carbon/glass HFRP bars demonstrating multi-stage stress-strain responses under tension, confirming ductility improvement through controlled fiber hybridization. Gao et al. [26] established optimal fiber ratios to prevent simultaneous fracture of hybrid fibers, while subsequent studies validated HFRP bar feasibility across strength grades [27] and proposed a novel constitutive model describing five-stage tensile behavior: linear elasticity, stress drop, recovery, plateau, and strain hardening [28]. Peng et al. [29] further corroborated ductility improvements through mesoscale finite element analysis. These collective findings substantiate HFRP bars' ductility superiority over conventional FRP bars, with established methodologies for determining critical fiber volume ratios.

Notably, the distribution pattern of fiber significantly influences composite tensile performance [30], while interlaminar shear failure has been identified as a critical failure mode in CFRP composites under transverse loading [31]. These insights confirm that shear effects during tensile loading, exacerbated by Poisson's effect and fiber alignment characteristics, disproportionately affect peripheral fibers. Consequently, the distribution pattern of fiber fundamentally governs the tensile performance of FRP bars, necessitating meticulous design considerations [32].

This study designed and produced three types of FRP bars: conventional glass fiber reinforced polymer (GFRP) bars, carbon fiber peripherally distributed hybrid FRP (H_1FRP) bars, and carbon fiber centrally concentrated hybrid FRP (H_2FRP) bars. Drawing on the fiber layer concept from composite laminates (Fig. 1), systematically experimental and numerical investigations were conducted to explore the influence mechanisms of fiber distribution on the tensile mechanical properties of HFRP bars. Based on visualized simulation results, elastic modulus correction parameters regarding fiber distribution patterns will be proposed. The investigation specifically elucidated the correlation between fiber distribution patterns and tensile performance in HFRP bars.

2. Experimental program

2.1. Tensile test

2.1.1. Preparation of HFRP bar

In this study, carbon and glass fibers were chosen to prepare HFRP bars. The HFRP bars required for the experiment were designed and prepared independently in the laboratory. In order to ensure the mechanical properties of HFRP bars, the total volume fraction of fiber was taken as 65 % and the volume fraction of resin was taken as 35 % [6]. To achieve the yield condition of the HFRP bar upon damage, it is necessary to ensure that the unloaded force upon damage of the low elongation fibers (carbon fibers) can be taken up by the remaining high elongation fibers (glass fibers). This requires limiting the critical doping of low elongation fibers, which is calculated by Eq. (1) to be 15 % for carbon fiber and 85 % for glass fibers in this study [25]. The required number of spools of glass fibers and carbon fibers are determined according to the calculated fiber volume and introduced continuously into the collector plate. After the steps of resin impregnation, orientation of the second layer of collector plate, pultrusion preform, winding, heat curing, cooling, and cutting according to the dimensions, two types of HFRP bars, the planned peripheral distribution of the carbon fibers

这种比例是为了实现伪延性行为，但是我们的比例，HF是无法承担LF（碳纤维）的卸载力。

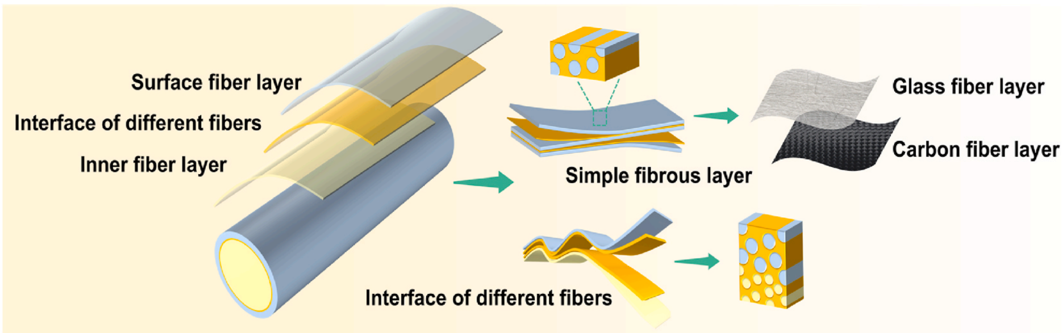

Fig. 1. Schematic diagram of "fiber layer" distribution in HFRP bars.

(H_1FRP) and the centralized distribution of the carbon fibers (H_2FRP), can be obtained.

$$\begin{cases} \dfrac{V_H}{V_L} \geq \dfrac{E_L \varepsilon_L}{E_H(\varepsilon_H - \varepsilon_L)} \\ V_H + V_L = 1 \end{cases} \tag{1}$$

where V_H and V_L respectively represent the volume fractions of fibers with high and low elongation rates; E_H and E_L respectively represent the Young's modulus of fibers with high and low elongation rates; ε_H and ε_L respectively represent the ultimate strain of fibers with high and low elongation rates.

The carbon fiber type selected in the preparation was T300, 12k, with a cross sectional area of the fiber bundle of 0.34 mm^2, and the glass fiber type selected was 2400 Tex, with a cross sectional area of the fiber bundle of 1.94 mm^2.Table 1 shows the mechanical properties of the materials used as provided by the manufacturer.

2.1.2. AE-based tensile testing of HFRP bars

The tensile tests were designed in accordance with ACI 440.3R-12 [33]. As shown in Fig. 2 (a), specimens with 1000 mm gauge length were prepared. Both ends were anchored using 300 mm-long steel tubes (2 mm wall thickness) filled with expansive cement to secure HFRP bars, preventing specimen damage by machine grips and eliminating initial defect impacts. Three specimen types were tested: carbon fiber peripherally-distributed bars (H_1FRP), carbon fiber centrally-concentrated bars (H_2FRP), and mono-glass fiber FRP bars. Three specimens were tested per group, with one specimen instrumented with piezoelectric sensors for acoustic emission (AE) monitoring (Fig. 2 (b)). The stress-strain curves were combined with AE signals to investigate fiber arrangement effects on tensile behavior mechanisms.

2.2. In-situ damage observation of fiber bundles via optical microscopy

In-situ observation of dynamic failure behavior in hybrid fiber bundles was conducted using a KEYENCE ultra-depth microscope system (VHX-7000) to verify the graded failure progression of hybrid fibers with varying elastic modulus. Identical displacement was applied to the hybrid fiber bundles through a displacement controller, enabling real-time observation of failure sequence and extent between carbon and glass fibers.

2.3. X-ray CT characterization

The specimens of the two types of HFRP bars after tensile damage were analyzed by three-dimensional imaging using X-ray computer tomography (X-ray CT, model: Vtomexs micrometer CT from GE), in order to quantitatively observe the damage morphology of the two types of HFRP bars after damage. The unit area after damage of carbon fiber and glass fiber was analyzed based on density difference and gray level. Test sample size 10 mm high, 10 mm diameter. The scanning voltage is set to 80 kV, the resolution is 9 μm, the cross section is derived by 0.05 mm spacing, and the longitudinal section is rotated by 1° interval.

2.4. SEM observation

The tensile damage at the interfaces of two types of HFRP bar fibers was observed using a Hitachi TM4000Plus scanning electron microscope equipped with energy dispersive spectrometer (EDS). The tensile damaged fiber bundles were collected and subjected to preparation processes such as cutting and gold spraying. A backscattering detector was selected for the test and the scanning voltage was set to 15 kV.

3. Results and discussion

3.1. Tensile test results

3.1.1. Stress-strain

The stress-strain curves obtained from the tensile tests of FRP bars are presented in Fig. 3(a)–(c). The H_1FRP bar demonstrated an ultimate tensile strength of approximately 983.3 MPa, which is 17 % lower than the 1185.33 MPa recorded for the H_2FRP bar. Compared with GFRP bars, the HFRP bars exhibited a noticeable stress recovery process after the initial failure occurs. This phenomenon is attributed to the graded failure mechanism of fibers with different elongation rates. Specifically, carbon fibers with lower

Table 1
Tensile property parameters of virgin materials.

Materials	Elasticity modulus (GPa)	Ultimate strength (MPa)	Ultimate strain (%)
Epoxy resin	45	55	1.1
Carbon fiber	239	4262	1.7
Glass fiber	81	2025	2.5

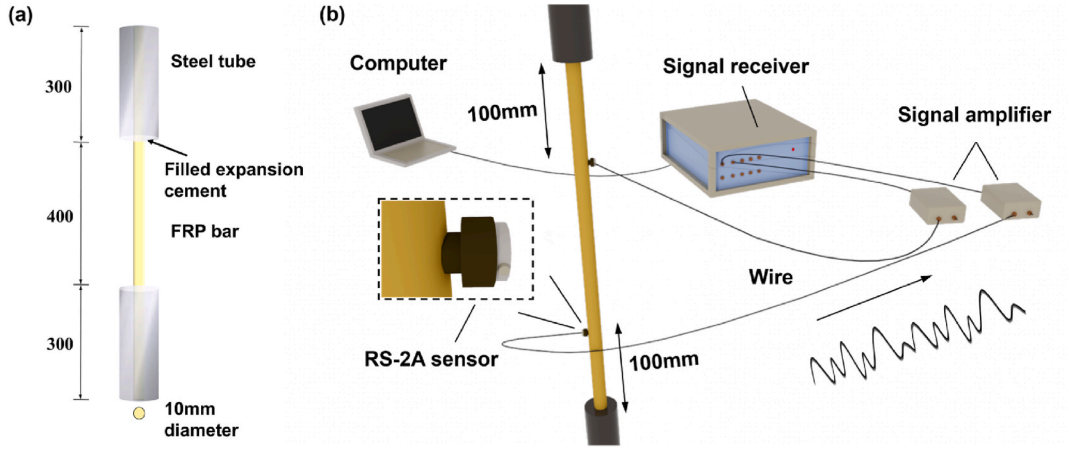

Fig. 2. Schematic diagram of the tensile specimen (a) Schematic diagram of the tensile specimen; (b) Schematic diagram of acoustic emission (AE) connection.

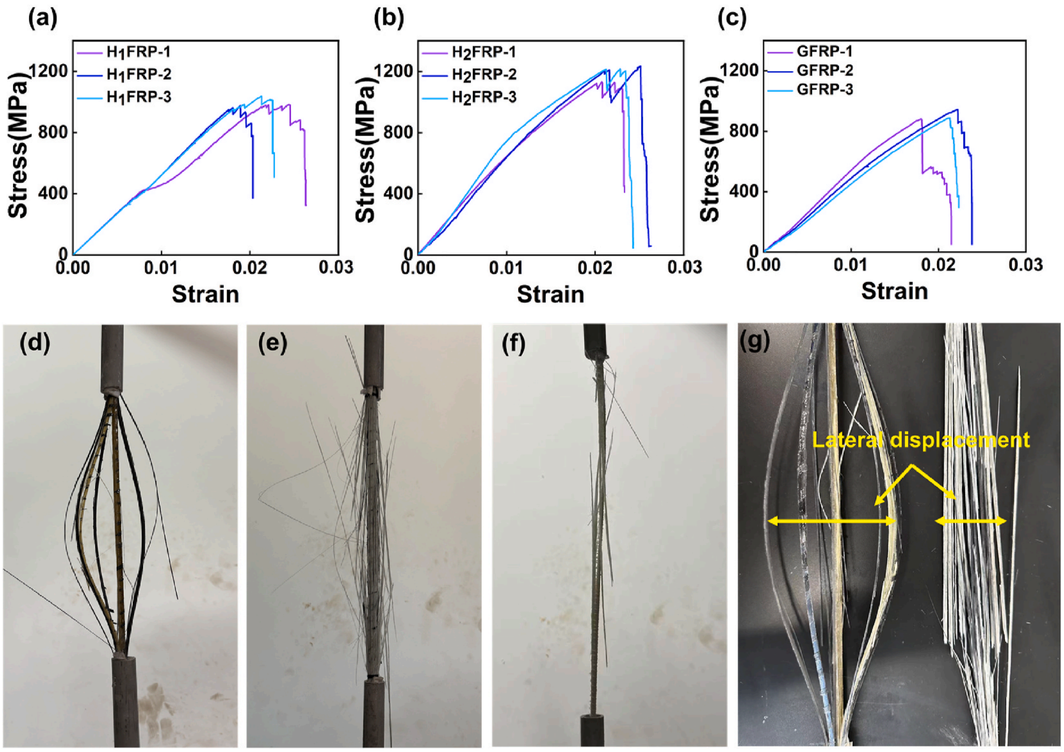

Fig. 3. Tensile test results and damage morphology of FRP bars. Stress-strain diagram of (a) H_1FRP bars; (b) H_2FRP bars; (c) GFRP bars; Tensile damage morphology of (d) H_1FRP bar; (e) H_2FRP bar; (f) GFRP bars; (g) Comparative Analysis of Tensile Damage in HFRP Bars.

elongation fail first, after which the unloaded stress is transferred to glass fibers with higher elongation through the shear stress of the resin matrix. Similar findings have been reported in Ref. [29]. In this study, experimental investigations were specifically designed and validated through relevant experiments to verify this graded failure mechanism of fibers with varying elongation rates, which will be comprehensively discussed in Section 3.2.

The fracture morphologies of the specimens under tensile failure are illustrated in Fig. 3(d)–(g). During tensile testing, the fiber bundles underwent tensile splitting and separation accompanied by an audible rupture. However, only an explosive cracking sound was detected during initial failure, with no visible surface damage observed on the FRP bars. This suggests that failure originated from inherent material imperfections. Comparing the failure morphologies of the three types of FRP bars in Fig. 3(d)–(f), it is not difficult to find that the failure of GFRP bars is accompanied by a large number of fiber collapses. The outer fibers were almost completely detached, and there was very little glass fiber left inside. This does not occur in HFRP bars. In addition, in Fig. 3 (g), pronounced fiber swelling was observed in externally wrapped carbon fibers, while minimal swelling occurred in internally embedded carbon fibers. This discrepancy arises from variations in the interfacial layer positions between carbon and glass fibers. Compared to the bonding strength within homogeneous fibers, the interfacial bonding between carbon and glass fibers exhibits weaker strength and greater susceptibility to positional distribution effects. A microscopic comparative analysis of interface layer damage will be conducted in Section 3.4 of this study.

Furthermore, as shown in Fig. 3 (d), for HFRP bars with carbon fibers peripherally distributed, incomplete fracture of carbon fibers occurred during failure, indicating insufficient interfacial bonding between carbon and glass fibers and underutilization of their ultimate strength. In contrast, carbon fibers located in the core region fractured almost completely, demonstrating full utilization of their ultimate strength. These observations align with the conclusions reported in Ref. [11].

3.1.2. Acoustic emission signal analysis

Damage signals generated during tensile failure processes can be utilized to analyze internal damage modes in HFRP bars with varying fiber distributions. Quantitative evaluation of damage type proportions provides critical insights into how fiber distribution patterns influence tensile performance characteristics. Acoustic emission (AE) waveforms constitute complex signals containing rich information. Classification of AE signal features reveals the temporal evolution of distinct damage modes. Key time-domain parameters include amplitude, energy, ring-down count, rise time, duration, peak frequency, and centroid frequency. Fig. S2 statistically analyzes the correlation of acoustic emission characteristics of two types of HFRP bars, determining peak frequency and rise time as the most appropriate clustering features, which is consistent with the results obtained from previous studies [34]. Optimal cluster number determination through combined analysis of silhouette coefficient and Davies-Bouldin Index (DBI) yielded three distinct clusters, corresponding to three damage mechanisms: fiber-resin debonding, resin cracking, and fiber fracture [34]. The accuracy of micro-damage clustering was verified in the supporting information. Inspired by the glass transition of resin at high temperatures, high-temperature heating of HFRP bars weakens the strength and bonding effect of the resin matrix. As shown in Fig. S1, after high-temperature heating, the damage signal energy representing resin cracking and fiber resin debonding decreases, while the energy released by fiber cracking does not change significantly. This confirms the accuracy of the clustering results.

The K-means clustering algorithm was employed to analyze AE signals. Raw data clustering features were processed using MATLAB software, where the built-in K-means function was implemented with silhouette coefficient evaluation for clustering quality assessment. After 50 iterations, the optimal clustering result was selected as the final output. Clustering distributions of AE signals for H_1FRP, H_2FRP, and GFRP bars are presented in Fig. 4(a)–(c), respectively.

The peak frequency of AE signals, defined as the frequency point with maximum energy in the spectrum, enables classification of internal material defects. As shown in Fig. 4, resin cracking signals predominantly exhibit peak frequencies in the 150–425 kHz range, while fiber-resin debonding signals concentrate within 75–150 kHz. Fiber fracture signals primarily demonstrate peak frequencies below 75 kHz.

Signal amplitude, indicative of event severity, shows positive correlation with damage intensity. Fig. 4 (a) reveals that 95 % of H_1FRP bar signals remain below 1000 dB (mean: 22.67 dB), with sporadic exceedances. In contrast, Fig. 4 (b) and (c) display significant increases in high-amplitude signals (>1000 dB constitutes 35 % and 48 % respectively), with mean amplitudes reaching 191.5

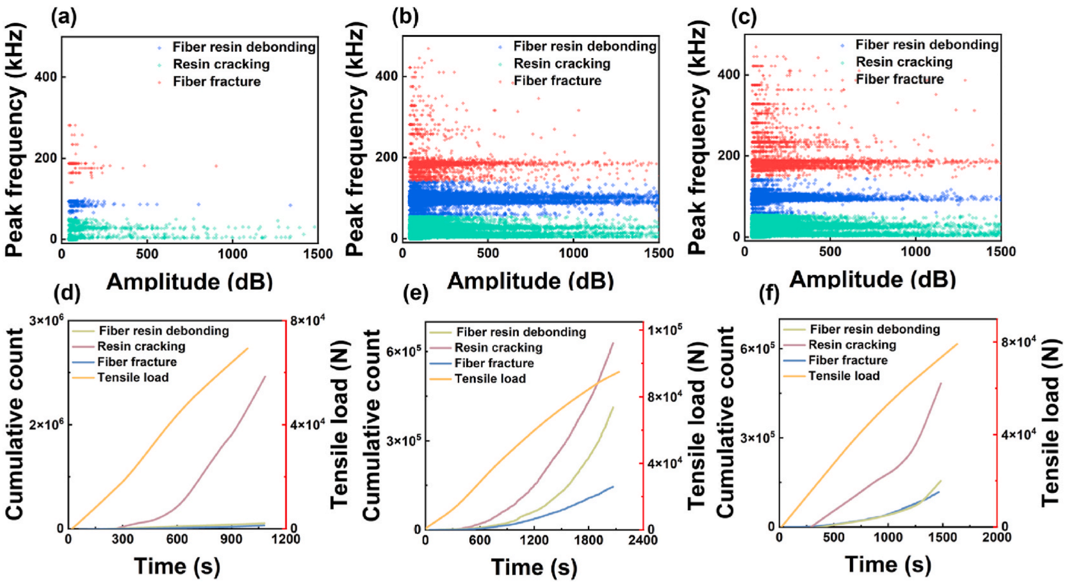

Fig. 4. Acoustic emission clustering results of (a) H_1FRP bar; (b) H_2FRP bar; (c) GFRP bar; Cumulative AE count response curve of (d) H_1FRP bar; (e) H_2FRP bar; (f) GFRP bar.

dB and 286.74 dB after normalization (Note: H_1FRP signal quantity represents 5 % of comparative groups). Experimental results demonstrate that H_2FRP bars exhibit pronounced resin cracking and interfacial debonding during linear elastic stage, accompanied by localized fiber fractures. Comparatively, H_1FRP bars show delayed fiber fracture initiation and reduced debonding occurrences. This discrepancy may be attributed to Poisson's ratio mismatch in hybrid fibers, which induces transverse displacement during tensile loading, thereby exacerbating internal damage progression. Additionally, carbon fibers with identical dimensions and cross-sectional shapes exhibit significantly higher stiffness than glass fibers. The peripheral distribution of carbon fibers effectively inhibits resin matrix cracking and interfacial debonding. In Section 3.3, matrix cracking behaviors between the two types of HFRP bar will be further compared through quantitative X-ray CT analysis.

Ring-down counts reflect signal intensity and activity. The cumulative counts for each failure mode were obtained by temporally accumulating ring-down counts, revealing the evolution patterns of different damage mechanisms during loading. Temporal variation curves of cumulative counts under three failure modes were plotted and compared with time-load curves, as shown in Fig. 4(e)–(g). In Fig. 4 (e), cumulative counts of fiber-resin debonding and fiber fracture showed no significant growth over time. However, Fig. 4 (f) and (g) demonstrate marked increases in cumulative counts during 500–800s of tensile loading, indicating multiple occurrences of fiber fracture and interfacial debonding within the bars. This observation confirms that the high stiffness of carbon fibers delays transverse damage progression in FRP bars, preventing premature fiber fracture and debonding during tension, thereby facilitating effective internal stress transfer. Comparative analysis of Fig. 4(f) and (g) reveals higher frequencies of fiber-resin debonding in carbon-doped H_2FRP bars compared to GFRP bars, potentially attributed to weak hybrid fiber interfaces and severe debonding effects. In Section 3.4, comparative SEM observations will be performed to investigate interfacial damage at fiber junctions.

3.2. *In-situ observation of tensile damage in fiber bundles via optical microscopy*

Existing studies generally recognize that the differential elastic modulus of hybrid fibers induce progressive failure during tensile loading, fundamentally explaining how fiber distribution patterns influence the mechanical properties of HFRP bars [28,29]. To validate this hypothesis, in-situ damage observations of hybrid fiber bundle fractures were conducted using the optical microscope integrated with EDS, with the damage process being video-recorded (the specific results refer to Video S1). Equal quantities of glass and carbon fiber bundles were selected for experimentation. Their ends were secured to a displacement controller and positioned under a super-depth optical microscope. Simultaneous displacement was applied to both fiber types, while the super-depth optical microscope dynamically recorded their fracture processes.

The experimental results are presented in Fig. 5, where Fig. 5(a)–(d) sequentially document four critical stages from initial state to fracture: relaxed state without displacement application, the pre-failure tension state, initial fracture of low-elongation carbon fibers, and subsequent damage initiation in high-elongation glass fibers. Experimental results confirmed that hybrid fibers exhibit progressive failure during tensile loading due to elongation discrepancies, with low-elongation fibers fracturing prior to their high-elongation counterparts. Notably, the fracture morphology of fiber bundles demonstrated a "lantern-shaped" configuration analogous to FRP bars. This observation confirms that the accompanied transverse displacement during tensile loading represents an inherent characteristic of fiber materials, thereby validating the hypothesis proposed in Section 3.1.2.

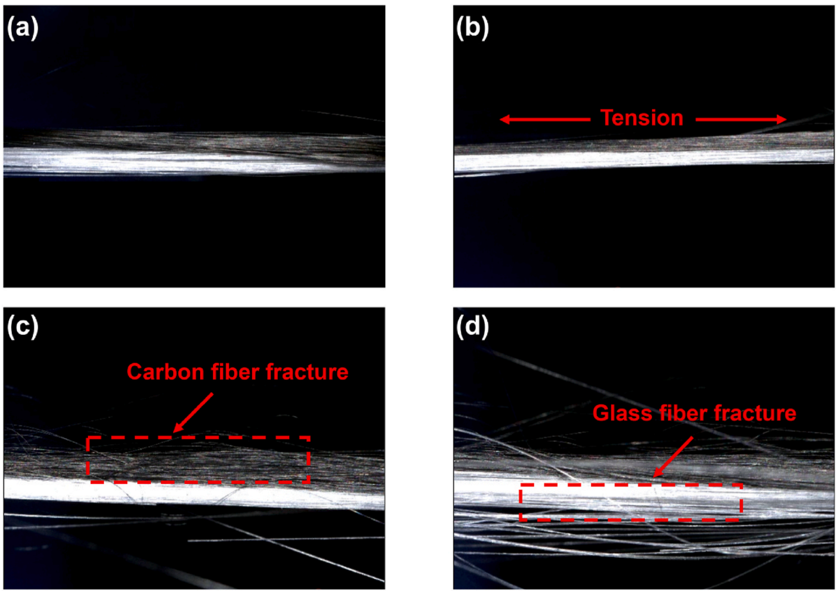

Fig. 5. Optical microscopy of hybrid fiber bundle tensile damage of (a) relaxed state without displacement application; (b) the pre-failure tension state; (c) initial fracture of low-elongation carbon fibers; (d) subsequent damage initiation in high-elongation glass fibers.

3.3. X-ray CT results

Gray-scale segmentation and quantitative analysis were performed on CT results of two HFRP bar types, with final outcomes illustrated in Fig. 6. As shown in Fig. 6 (a), external fibers exhibited higher damage levels than internal fibers in both HFRP bars after failure. Compared to H_1FRP bars, H_2FRP bars demonstrated more severe fiber-resin debonding, greater fiber bundle separation, and increased internal crack formation. Original coarse fiber bundles were fragmented into dense fine bundles. The interfacial regions between carbon and glass fibers existed as filamentous states, indicating aggravated interfacial damage. In contrast, H_1FRP bars retained relatively intact coarse fiber bundles with fewer internal cracks in carbon and glass fibers, though significant fiber-resin debonding was observed. Notably, the interfacial zones in H_1FRP bars displayed smooth and complete separation between carbon and glass fiber layers. These experimental results align with conclusions from acoustic emission tests in Section 3.1.2., we propose that transverse shear forces influence FRP bars during tensile loading. As depicted in Fig. 6 (b), these transverse shear forces likely originate from Poisson effects and relative displacement differences between high and low elongation fibers during stretching. The impact of transverse shear forces during tensile processes will be further investigated through simulation studies in Section IV.

Fig. 6(c)–(e) present quantitative analysis of specific surface area in HFRP bars. The results revealed a gradual increase in fiber bundle specific surface area from the surface to the core of HFRP bars. This phenomenon suggests that transverse shear forces induce discontinuous damage progression in lateral fiber bundles. Significant differences in damage extent were observed between carbon (glass) fibers distributed in the core and outer regions. In H_1FRP bars, carbon fibers at outer regions exhibited specific surface area concentrated within 0–5 % range, whereas in H_2FRP bars, core-distributed carbon fibers showed minimum and maximum specific surface area values ranging from 25 % to 36 %.

Fiber volume distribution data for failed bars are compiled in Fig. 7. HFRP bars fabricated through resin bonding of unidirectional fibers, exhibit insufficient strength in the thickness direction and are prone to delamination under tensile loading [34,35]. Consequently, quantitative analysis was performed on both the outer and inner fiber layers in this study. Comparative analysis of identical fiber types in Fig. 7(e) and (f) versus Fig. 7(g) and (h) reveals significant damage variation depending on fiber distribution positions. Outer-layer fibers exhibited substantially higher damage levels than their inner-layer counterparts. This damage discrepancy between external and internal regions correlates closely with transverse shear force effects during tension.

Fig. 7(c)–(h) present fiber volume retention rates and average bundle volumes in two HFRP bar types post-failure. The results indicate higher volume retention rates for inner-distributed fibers ($R_{G1} = 97.22\%$, $R_{C2} = 93.87\%$) with weaker fiber-resin interfacial debonding and lower damage levels, while outer-distributed fibers showed reduced retention rates ($R_{C1} = 81.93\%$, $R_{G2} = 82.73\%$) accompanied by stronger interfacial debonding and severe damage. Notably, glass fibers exhibited higher volume retention than carbon fibers in both HFRP types, attributed to premature failure of low-elongation carbon fibers during tensile loading followed by progressive volume loss. Comparative analysis of average fiber bundle volumes revealed that H_1FRP bars contained larger bundles ($V_{G1} = 108.22mm^3$, $V_{C1} = 12.07mm^3$) compared to H_2FRP bars ($V_{G2} = 30.68mm^3$, $V_{C2} = 23.21mm^3$). It indicates that carbon fiber

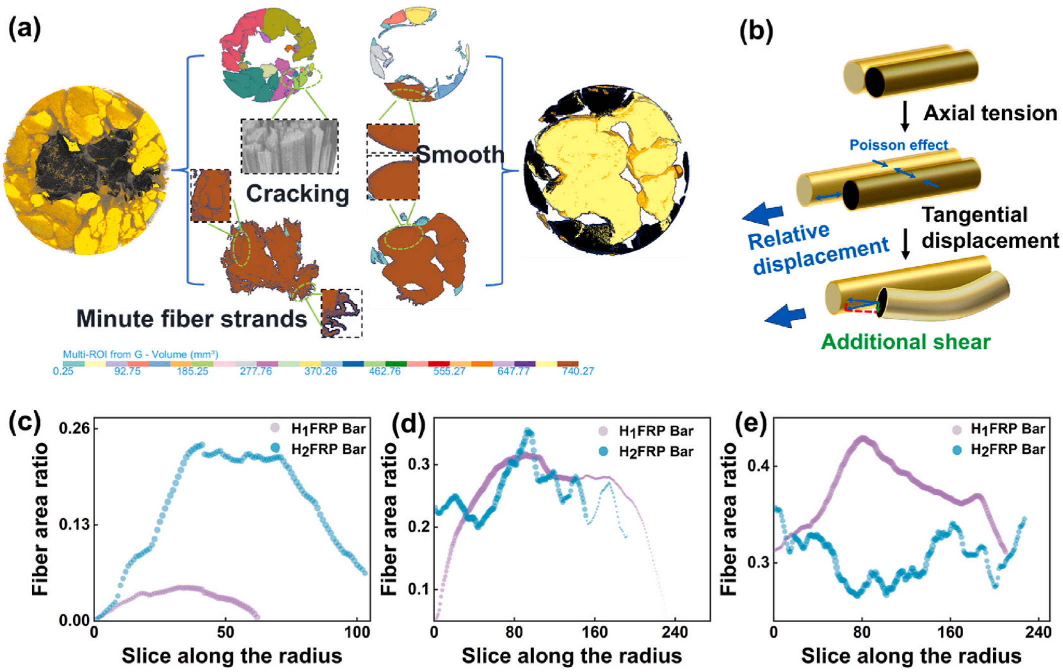

Fig. 6. Quantitative analysis of X-ray CT transverse slices (a) Damage morphology of transverse sections; (b) Poisson effect-induced transverse shear; Fiber diameters transverse distribution of (c) Edge distance interval from the bar: [0, 1.5); (d) Edge distance interval from the bar: [1.5, 3); (e) Edge distance interval from the bar: [3, 5).

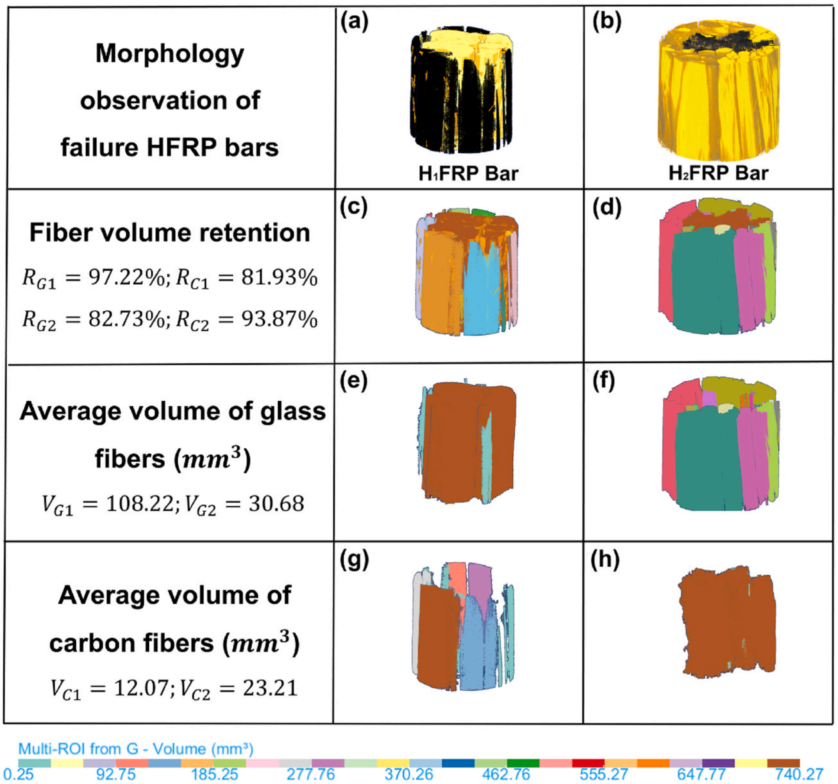

Fig. 7. Longitudinal quantitative analysis of X-ray CT Morphology observation of (a) failure H_1FRP bars (b) failure H_2FRP bars; Fiber volume retention of (c) H_1FRP bars; (d) H_2FRP bars; Average volume of glass fibers of (e) H_1FRP bars; (f) H_2FRP bars; Average volume of carbon fibers of (g) H_1FRP bars; (h) H_2FRP bars.

concentrated distributed in H_2FRP resulted in severe fiber separation, more serious matrix cracking, and intensified fiber-resin debonding, which is consistent with acoustic emission testing observations.

3.4. SEM results

The fiber-resin interfacial bonding is critical for inter-fiber force transmission, with its microscopic damage morphology directly governing the mechanical properties of HFRP bars [36,37]. Distinct elastic modulus between carbon and glass fibers creates vulnerable

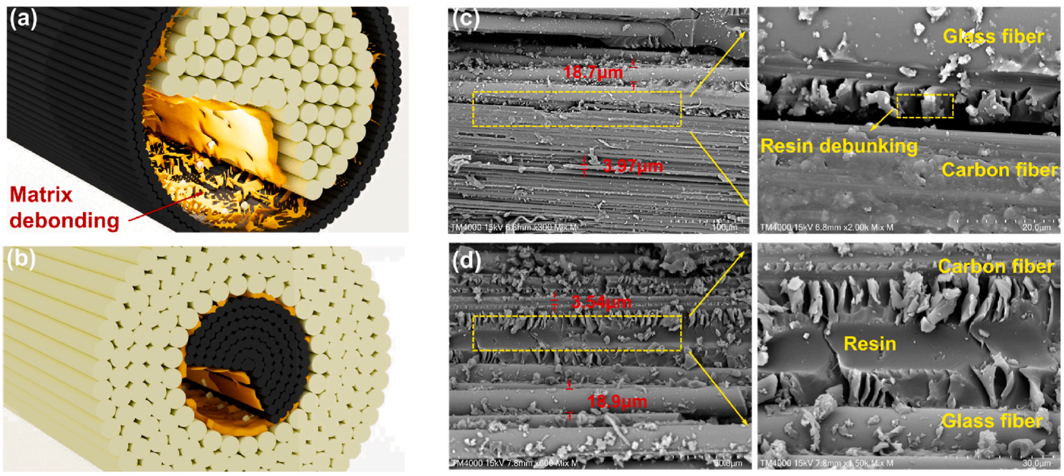

Fig. 8. SEM characterization results: Interface damage diagram of (a) H_1FRP bar; (b) H_2FRP bar; SEM characterization of interfacial damage of (c) H_1FRP bar; (d) H_2FRP bar.

zones at their contact interfaces, inducing intensified stress concentration. The damage morphology at the different fiber interface of the two types of HFRP bars is shown in Fig. 8. The thicker fiber is glass fiber (approximately 18–19 μm) and the thinner one is carbon fiber (approximately 3–4 μm). As illustrated in Fig. 8(a) and (b), distinct interfacial positions result in markedly divergent damage morphologies under tensile failure.

Fig. 8 (c) and (d) comparatively present the damage morphologies at fiber interfaces between H_1FRP and H_2FRP bars under SEM observations. In H_1FRP bars, interfacial regions near the periphery exhibit pronounced resin debonding and distinct fiber layer

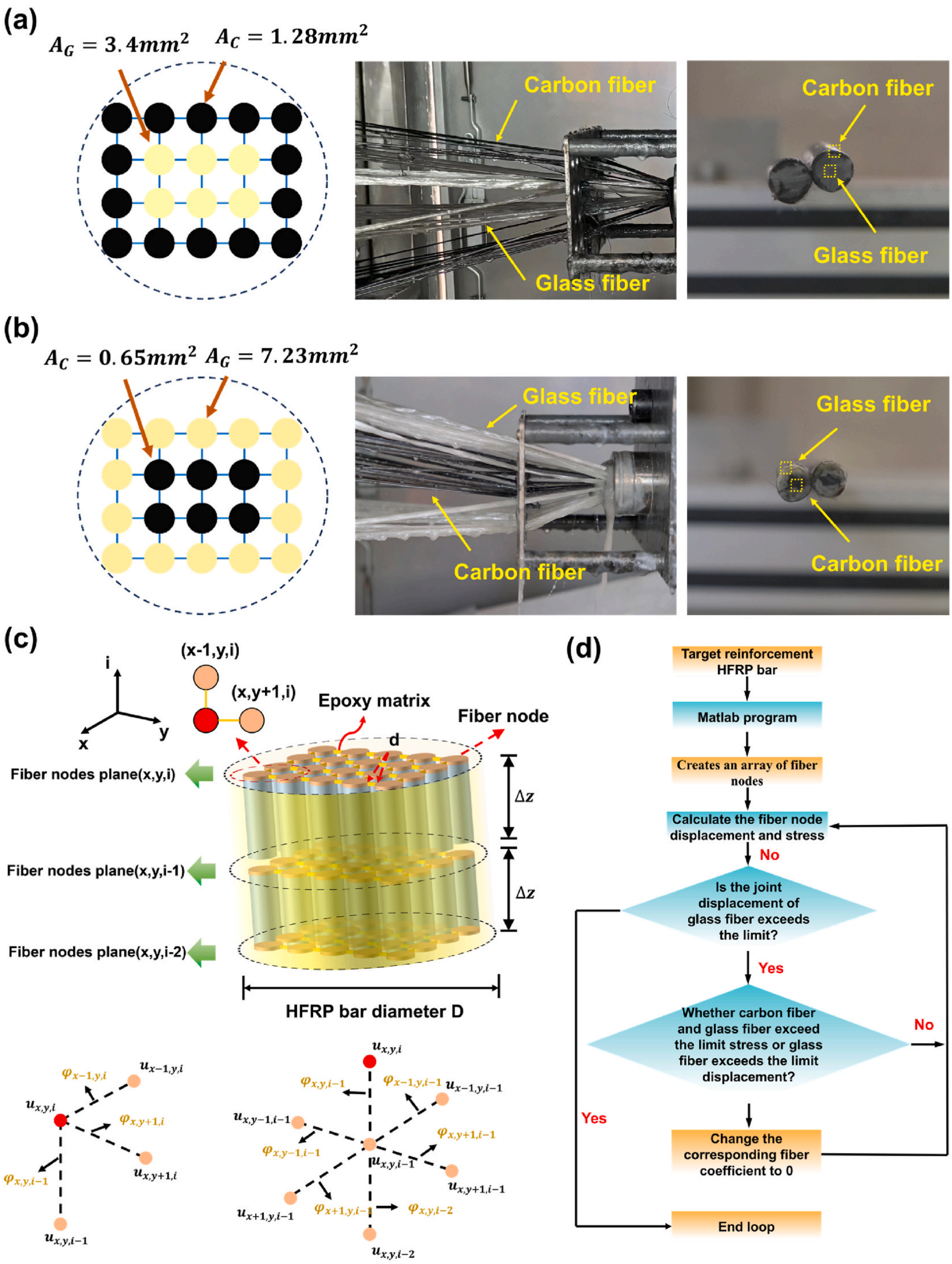

Fig. 9. Finite element model developed based on production requirements for resin simulation analysis: Parametric design basis of planar configuration of (a) H_1FRP bar; (b) H_2FRP bar; (c) 3D spatial schematic of the model; (d) Computational flow chart of numerical procedure.

separation. The reason is outer-distributed carbon fibers form dispersed bundles, establishing extensive contact areas with glass fiber layers where the contact interface becomes more vulnerable compared to carbon fiber-concentrated distributions. Enhanced shear effects exacerbate interfacial debonding. Conversely, H_2FRP bars position the high-low elongation fiber interface internally, minimizing transverse shear impacts. Compared with H_1FRP bars, H_2FRP bars demonstrate better-preserved resin adhesion between the two types of fibers, characterized by weaker interfacial damage. This configuration facilitates more effective stress transfer between fibers of differing elongation capacities.

4. Numerical simulation

Based on the experimental findings, it is evident that investigating the influence of fiber distribution patterns on the tensile performance of HFRP bars requires initial clarification of shear effects during tensile test [15,29]. Shear forces are primarily transmitted through the resin matrix in HFRP bars. The load released from fractured fibers is transferred via shear to adjacent fiber bundles during tensile test, thereby maintaining structural stability [29]. In this context, a local load-sharing (LLS) rule was adopted to develop finite element models for characterizing stress transfer mechanisms in HFRP bars under tension [29]. The model also accounts for the resin matrix's contribution to longitudinal tensile resistance [22,38].

This study extends an existing three-dimensional shear-lag model by incorporating experimental parameters (as shown in Fig. 9 (a)–(b)) to numerically simulate realistic damage evolution in fiber units during tension [39]. The distribution of fiber elements was simulated using a three-dimensional matrix, and fibers are regarded as rod units while the resin matrix is regarded as spring units. Stress and displacement distributions across fiber units under varying strain increments were obtained after model convergence.

4.1. Basic theory

Prior investigations have demonstrated that a minimum of five fiber elements are required to replicate fiber-reinforced composite behaviors [40]. Accordingly, the finite element analysis model assumes quadrilateral configuration of fiber alignment, as illustrated in Fig. 9 (a)–(b). The cross-section fiber bundle arrangement adopts a 4 × 5 rectangular array, with each fiber longitudinally discretized into ten elements. Fig. 9 (c) presents full-scale modeling matching experimental parameters: individual fiber bundles measure 400 mm in length (consistent with HFRP bar tensile specimens), comprising 20 fibers. Each fiber is discretized into 10 elements with 40-mm unit length, and each unit is regarded as rod unit. Based on shear-lag theory, micromechanical equilibrium equations are formulated for specified fiber nodes (x, y, z) [39].

$$A\frac{d\sigma_{x,y,z}(z)}{dz} = -\frac{\pi}{4}r\sum_{l=1}^{p}\tau_{x,y,z}^{l} \tag{2}$$

where A represents the cross-sectional area of fibers; $\sigma_{x,y,z}(z)$ represents the axial stress of fiber elements(x,y,z); r represents the radius of fiber elements; p represents the total number of adjacent fibers in the plane; $\tau_{x,y,z}^{l}$ is the shear stress for the neighboring node.

Assuming perfect bonding at fiber element nodes, the resin matrix is regarded as a spring unit. The governing differential equations for fiber tensile stress and matrix-mediated shear stress transfer are derived using the finite difference method:

$$\frac{d\sigma_{x,y,z}(z)}{dz} = \frac{E_f}{L^2}\left\{\gamma_{x,y,z+1}\left[u_{x,y,z+1} - u_{x,y,z}\right] + \gamma_{x,y,z-1}\left[u_{x,y,z} - u_{x,y,z-1}\right]\right\} \tag{3}$$

$$\tau_{x,y,z}^{l} = \frac{G_m}{d}\left[u_{x',y',z} - u_{x,y,z}\right] \tag{4}$$

where E_f represents the Young's modulus of fibers; L represents the length of fiber elements; $\gamma_{x,y,z+1}$ is set to zero if the fiber element between node (x,y,z) and (x,y,z+1) is broken, otherwise it takes 1; $u_{x,y,z+1}$ represents the displacement of fiber elements between node (x,y,z) and (x,y,z+1); G_m represents the Young's modulus of resin; $u_{x',y',z}$ represents the displacement of fiber elements surrounding node (x,y,z) in cross section.

Substitution of Equations (3) and (4) into Equation (2) yields the iterative equation governing fiber nodal displacements:

$$\begin{aligned}&\frac{AE_f}{\Delta z^2}\left[\gamma_{x,y,z+1}\left(u_{x,y,z+1} - u_{x,y,z}\right) + \gamma_{x,y,z}\left(u_{x,y,z} - u_{x,y,z-1}\right)\right] + \\ &\frac{\pi}{4}r\sum_{p}^{l=1}\left[G_m\frac{u_{x',y',z} - u_{x,y,z}}{d}\left(1-{\varphi_l}^2\right)\left(1-\mu^2\right) + \left(1-{\varphi_l}^2\right)\tau_s\right] = 0, (2 \le z \le N+1)\end{aligned} \tag{5}$$

where p is the total number of adjacent fibers in the cross section; τ_s is the residual shear stress after the failure of the resin matrix; E_f represents the Young's modulus of fibers, $\gamma_{x,y,z+1}$ determines the damage between the nodes (x,y,z) and (x,y,z+1), and the initial value of $\gamma_{x,y,z+1}$ is 1, and the value becomes 0 when the fiber at (x,y,z+1) fails; G_m denotes the shear modulus of the matrix; φ_l is the damage coefficient, with an initial value of 1, which becomes 0 when the corresponding fiber unit is damaged; μ denotes elastic coefficient $\mu = 1$ indicates the elasticity of the resin, and $\mu = 0$ indicates that the resin is in a plastic state; d is the distance between fiber bundles and $u_{x,y,z}$ is the displacement of node (x,y,z); the relevant basic geometric parameters of the model are shown in Table 2.

In commercially produced HFRP bars, the fiber volume fraction is maintained at 65 %, comprising 15 % carbon fibers and 85 % glass fibers. The simulation model replicates the actual fiber bundle distribution patterns. As illustrated in Fig. 9(a), the externally wrapped carbon fiber configuration employs 6 carbon fiber bundles and 14 glass fiber bundles (total $N_f = 20$ bundles), where the cross-sectional area per orifice unit measures $A_C = 1.28mm^2$ for carbon fibers and $A_G = 3.4mm^2$ for glass fibers. This configuration corresponds to industrial practice where quadruple carbon fiber bundles penetrate single holes, while double glass fiber bundles occupy individual holes. Conversely, as illustrated in Fig. 9(b), the internally concentrated carbon fiber configuration utilizes 14 carbon fiber bundles and 6 glass fiber bundles (total $N_f = 20$ bundles), with $A_C = 0.65mm^2$ and $A_G = 7.23mm^2$ respectively, equivalent to double carbon fiber bundles per hole versus quadruple glass fiber bundles. The analysis assumes perfect interfacial bonding without resin matrix participation in axial load-bearing. Epoxy resin parameters are determined through manufacturer specifications and tensile test data, with scale and shape parameters set at 50 MPa and 10 respectively. Geometric parameters of hole units are systematically presented in Table 3.

4.2. Calculation process

The random strength allocation of fiber nodes is achieved by using the Weibull distribution. The carbon fiber bundles are assumed with shape parameter 4200 and scale parameter 50, while the glass fiber bundles are assigned shape parameter 2000 and scale parameter 30. The failure stress of fiber bundle elements is determined through Equation (6):

$$\sigma^s_{x,y,z} = \left[ln\left(\frac{1}{1-P_f}\right)\frac{L_0}{\Delta z}\right]^{\frac{1}{\rho}}\sigma_0 \tag{6}$$

where $\sigma^s_{x,y,z}$ denotes the ultimate strength of fiber notes with coordinates of (x,y,z); P_f denotes the failure probability of fibers; L_0 denotes the scale parameter; Δz denotes the length of each fiber unites; ρ denotes the shape parameter; σ_0 denotes the tensile strength of fibers(shown in Table 1).

Fig. 9 (d) displays the computational flowchart of the numerical model, with implementation achieved through MATLAB R2024a. Array models were separately constructed for H_1FRP and H_2FRP bars. Following the LLS rule, load transfer between fiber nodes in the matrix occurs exclusively through shear deformation, while the influence of remote nodes is disregarded. Parameter adjustments were performed to simulate carbon fiber distributions in peripheral and concentrated configurations. Initial array configurations were derived from experimentally measured fiber bundle strengths and ultimate tensile strains. The tensile loading sequence was replicated through controlled strain increments. Each iteration required verification of fiber displacement and stress fields. Upon detecting stress exceedance in any fiber element or carbon fiber strain surpassing thresholds, the corresponding axial force coefficient was reset to zero (denoting element deactivation) with cycle reiteration. Glass fiber displacement exceeding tolerance limits triggered computational termination. This control condition is derived from the fracture sequence of high and low elongation fibers and has been experimentally verified in Section 3.2. This strain-controlled framework reproduces the axial force transfer mechanism from fractured low-elongation fibers to intact high-elongation fibers via matrix shear interaction.

The iterative procedure incorporates progressive strain updates (ε) with 0.03 strain increments per cycle, accompanied by boundary condition updates. Here, the 0th fiber unit is regarded as the fixed end, while the *Nth* fiber unit is regarded as the traction end. When the displacement of the traction end reaches the ultimate tensile displacement of the FRP bar, the simulation program will end the iteration. The specific boundary conditions are as follows:

$$u_{x,y,1} = 0, u_{x,y,N+1} = \varepsilon NL \tag{7}$$

where N is the number of longitudinal fiber units, L is the length of fiber units, and $N+1$ refers to the $N+1$st layer of planar fiber nodes; $\frac{AE_f}{\Delta z^2} = a$ (a is a constant), $\frac{\pi}{4} r = b$ (b is a constant). The above formula can be simplified as:

$$u_{x,y,i} = \frac{B \times u_{x,y,i-1} - C \times u_{x,y,i-2} - D - E \times F}{A} \tag{8}$$

where $A = a\varphi_{x,y,i}$; $B = a(\varphi_{x,y,i} + \varphi_{x,y,i-1}) + b\frac{G_m}{d}\sum \varphi_{x',y',i-1}$; $C = a\varphi_{x,y,i-1}$; $D = b\tau_r(n - \sum \varphi_{x',y',i-1})$; ($n$ is the number of neighboring nodes of the computed node) $E = b\frac{G_m}{d}$; $F = \sum \varphi_{x',y',i-1} u_{x',y',i-1}$;

Based on the shear lag equations with the influence of remote fiber nodes neglected, fiber displacements exhibit strong dependence on adjacent nodal displacements. Consequently, the determination of parameters A-F must be based on the specific locations of

Table 2
Design geometric parameters of the model.

Item	Size
Fiber spacing d	0.42 mm
Model diameter D	10 mm
Fiber length Δz	40 mm
Number of fibers N_f	20

Table 3
Geometric parameters of finite element model.

Pore element	Actual hole area (mm^2)	Model hole area (mm^2)	Relative error (%)
H_1FRP Glass fiber	3.88	3.4	−12.3
H_1FRP Carbon fiber	1.36	1.28	−5.88
H_2FRP Glass fiber	7.76	7.23	−6.83
H_2FRP Carbon fiber	0.68	0.65	−4.41

investigated fiber nodes. Appendix Table 1 systematically presents the parameter values corresponding to various fiber positions under investigation.

Upon convergence of nodal displacement iterations, the axial tensile stress in fiber elements can be determined as:

$$\sigma_{x,y,z} = E_f \frac{u_{x,y,z+1} - u_{x,y,z}}{\Delta z} \tag{9}$$

where $\sigma_{x,y,z}$ denotes the failure strength of fiber with coordinate of (x,y,z) based on simulation results; E_f denotes the Young's modulus of fibers.

Subsequently, the average stress of fiber elements can be calculated as:

$$\sigma_b = \frac{1}{N \times M} \sum_{x=1}^{M} \sum_{y=1}^{N} \sigma_{x,y,z} \tag{10}$$

where σ_b denotes the average failure strength of fibers; N is the number of longitudinal fiber units; N is the number of cross-sectional fiber units.

4.3. Numerical simulation results

Fifty numerical simulations were conducted on HFRP bars with distinct carbon fiber distribution patterns. Stress and strain data at fiber nodes were systematically recorded and averaged for analysis. Fig. 10 (a)–(b) presents post-elastic phase simulation results,

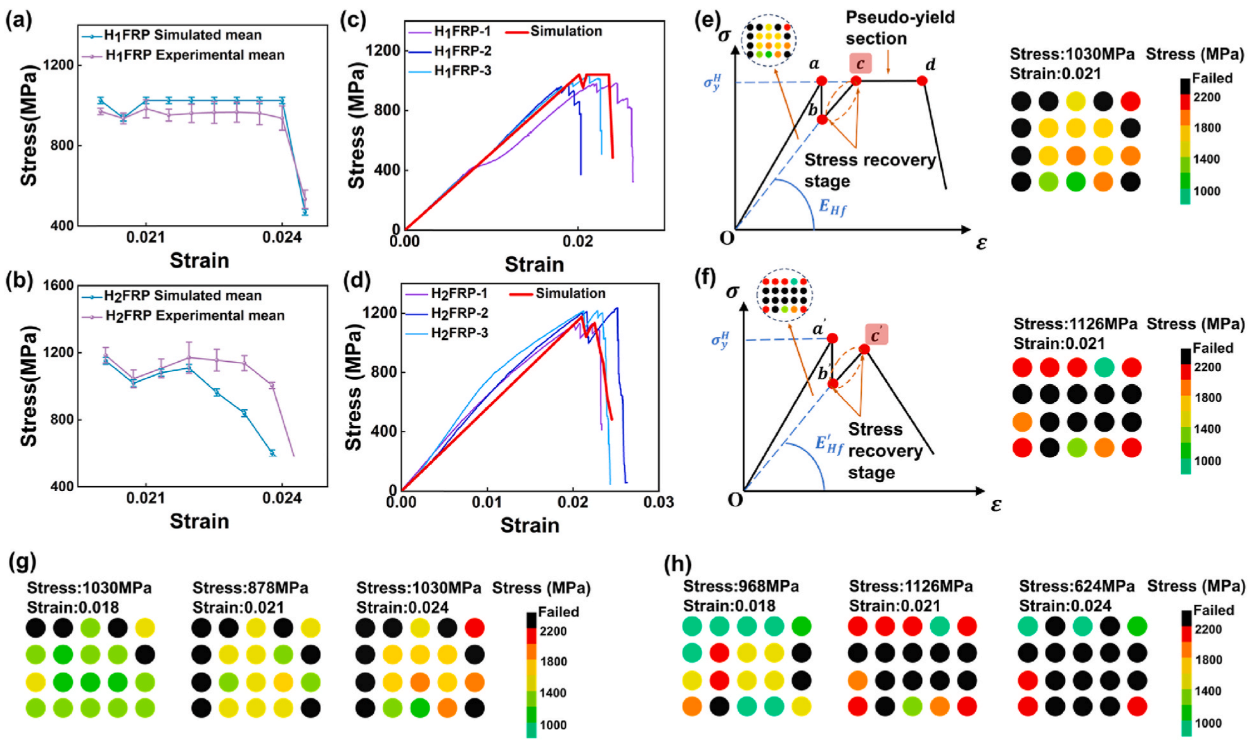

Fig. 10. Numerical simulation results: Comparative analysis under equivalent strain increments (ad/a' c') of (a) H_1FRP bar; (b) H_2FRP bar; Full-process experimental-numerical comparison of (c) H_1FRP bar; (d) H_2FRP bar; Stress distribution at fiber nodes during stress recovery phase termination of (e) H_1FRP bar; (f) H_2FRP bar; Partial moment stress distributions illustrating stress reduction to recovery transitions of (g) H_1FRP bar; (h) H_2FRP bar.

revealing consistent stress evolution trends between experimental and numerical outcomes under equivalent strain increments. The simulated tensile strength of H_1FRP bars reached 1024.3 MPa, representing a 12.4 % reduction compared to H_2FRP bars (1151.67 MPa). This numerical discrepancy demonstrates consistency with experimental findings obtained from tensile tests. Numerical results in Fig. 10 (c)–(d) demonstrate stress recovery phases during failure for both bar types, with HFRP bars containing peripherally distributed carbon fibers exhibiting more pronounced yielding characteristics.

Stress degradation in HFRP bars was observed during carbon fiber fracture due to uneven interfacial stress transfer between carbon and glass fiber layers. Fig. 10 (g) illustrates stress variations in fiber nodes under different strain levels. Severe resin debonding at the carbon-glass fiber interface in H_1FRP bars (as revealed by SEM analysis in Section 3.4) caused incomplete load transfer from damaged carbon fibers to internal glass fibers after initial carbon fiber failure. Only partial stress redistribution occurred through adjacent glass fibers receiving residual loads from fractured carbon fibers. This insufficient fiber strength utilization mechanism reduced the ultimate tensile strength of HFRP bars, with H_1FRP specimens exhibiting approximately 12 % lower strength than H_2FRP counterparts. Progressive displacement accumulation gradually deactivated carbon fibers, while internal glass fiber layers demonstrated effective stress redistribution due to minimal shear interference and well-preserved fiber-resin interfaces, ultimately manifesting an extended pseudo-yield stage.

In contrast, HFRP bars with concentrated carbon fiber distribution (Fig. 10 (h)) exhibited uniform carbon fiber damage during failure, accompanied by robust resin bonding at carbon/glass fiber interface transition layers (as evidenced by SEM observations in Section 3.4). The unloading stress was effectively transferred through resin shear forces to adjacent glass fibers, enhancing the composite's overall load-bearing capacity. However, glass fiber layers distributed at the bar periphery (subjected to significant shear forces) demonstrated severe fiber-resin debonding (quantitative X-ray CT analysis in Section 3.3). As carbon fibers gradually deactivated, partial glass fiber failure induced by transverse shear forces resulted in inadequate stress redistribution and limited ductility enhancement, ultimately leading to abrupt structural failure.

4.4. General discussion

The simulated results demonstrated substantial agreement with experimental observations. Overall, H_2FRP bars exhibited lower yield ductility compared to H_1FRP bars, while H_1FRP bars showed inferior ultimate tensile strength relative to H_2FRP specimens. Both HFRP variants displayed notable stress recovery stages, significantly improving tensile brittleness. Notably, experimental stress-strain curves revealed oscillatory characteristics during the "pseudo-yield" phase of HFRP bars. As shown in Fig. 10 (a), normalized strain scaling demonstrated consistent variation trends between simulations and experiments, suggesting these oscillations correspond to internal stress redistribution processes. Although numerical simulations could only calculate stress at strain increments without capturing intra-step variations as experimental equipment does, the mesoscale mechanical analysis successfully revealed the differential effects of fiber distribution patterns on stress redistribution mechanisms.

5. Modification of tensile constitutive prediction model based on fibers distribution

Current constitutive models for predicting HFRP bar tensile behavior primarily include three-phase [23] and five-phase [28] formulations, both neglecting fiber distribution pattern effects. This study reveals that both the spatial distribution of unidirectional fiber layers and interfacial configurations between hybrid fibers significantly influence tensile damage mechanisms in HFRP bars. Experimental and numerical analyses demonstrate that fiber distribution patterns primarily affect tensile performance through shear effect during loading processes. Given the strong correlation between simulations and experimental results, the progressive damage visualization derived from numerical models enables refinement of the stress recovery phase in existing five-phase constitutive models. A modified five-phase constitutive model was developed, incorporating fiber distribution characteristics based on simulated nodal damage patterns to enhance predictive accuracy.

5.1. Formula modification

5.1.1. $\boldsymbol{ab}/\ \boldsymbol{a'b'}$ segments

As illustrated in Fig. 10 (e)–(f), a modified constitutive model incorporating fiber distribution configurations is proposed. Elastic modulus correction factors are recommended according to fiber bundle failure patterns observed during stress recovery phases in numerical simulations. Both H_1FRP and H_2FRP bars exhibit three characteristic phases: elastic phase (Oa/Oa'), abrupt stress reduction phase ($ab/a'b'$), and stress recovery phase ($bc/\ b'c'$). The critical distinction lies in the post-recovery stress redistribution stability (pseudo-yield phase).

Notably, load transfer from carbon to glass fibers occurs predominantly through shear mechanisms. Shear effects remain negligible until carbon fiber failure occurs. Consequently, existing five-phase models [27] remain applicable for designing the elastic ($Oa/\ Oa'$) and abrupt stress reduction ($ab/a'b'$) phases of both HFRP types, which are not the primary focus of this investigation.

5.1.2. $\boldsymbol{bc}/\boldsymbol{b'c'}$ segments

Beyond Point b, the differential lateral deformation of carbon fibers induced by internal shear stress variations leads to distinct constitutive responses between peripherally distributed and centrally concentrated fiber configurations [30,31]. As illustrated in Fig. 10(e) and (f), all HFRP bars exhibit a stress recovery phase ($bc/\ b'c'$ segments) following initial stress decline. Numerical simulations reveal that the initial stress reduction primarily stems from low-elongation fibers exceeding their ultimate strain and ceasing load-bearing.

Subsequently, the redistributed stress from failed carbon fibers transfers through shear mechanisms to adjacent active fibers, enabling continued load-bearing through surviving fibers. Notably, the reduced slope during stress recovery compared to the *ab* segment necessitates introducing E_{Hf} to characterize the effective elastic modulus in this phase. Following the rule of mixtures, E_{Hf} is determined by the modulus of residual active fibers (resin matrix contribution excluded [28]), derived as:

$$E_{Hf} = V_f'\left(V_L'E_L + V_HE_H\right) \tag{11}$$

where V_f' is the total volume fraction of remaining working fibers and V_L' is the volume fraction of remaining working low elongation fibers. With reference to the simulation results, suggested values of correction factors α、β are given to calculate V_f' and V_L':

$$\begin{aligned} V_f' &= \alpha V_f \\ V_L' &= \beta V_L \end{aligned} \tag{12}$$

When the carbon fiber is outsourcing distribution, $\alpha_1 = 0.92, \beta_1 = 0.357$; when the carbon fiber is centralized distribution, $\alpha_2 = 0.85$, $\beta_2 = 0$ (i.e., all the carbon fibers quit working).

For simplified calculation, equal strain values at Points *a* and *b* are assumed [28]. A secant line with slope E_{Hf} is constructed through the origin, intersecting the vertical extension of Point *a* to determine the stress-strain coordinates of Point *b*.

For centrally concentrated carbon fiber configurations, point *c'* defines the critical failure position of carbon/glass hybrid FRP bars under tensile loading. The ultimate tensile strain at this position is governed by the high-elongation fiber bundles [28]:

$$\varepsilon_{c'} = \varepsilon_H \tag{13}$$

where $\varepsilon_{c'}$ represents the ultimate tensile strain of the HFRP reinforcement at point c', thus the ultimate stress at point c' ($\sigma_{c'}$) can be obtained as:

$$\sigma_{c'} = E_{Hf}{}'\varepsilon_{c'} = \alpha_2 V_f(\beta_2 V_LE_L + V_HE_H)\varepsilon_{c'} \tag{14}$$

For HFRP bars with peripherally distributed carbon fibers, point *c* does not represent the terminal failure position. Assuming stress recovery in segment *bc* reaches the material's ultimate strength ($\sigma_c = \sigma_a$), a secant line with modified modulus E_{Hf} is constructed through the origin. This line intersects the $\sigma = \sigma_a$ extension at Point *c*, thereby determining its stress-strain coordinates, where:

$$E_{Hf} = \alpha_1 V_f(\beta_1 V_LE_L + V_HE_H) \tag{15}$$

5.1.3. *cd* segments

HFRP bars with peripheral carbon fiber distribution exhibit a pseudo-yielding phase post-stress recovery, characterized by stress fluctuations within a defined range during strain progression. This phenomenon manifests as stress "oscillations" in HFRP bars despite continuous strain increases. Numerical simulations in Section IV reveal that this phase primarily involves stress redistribution among residual working fibers after reattaining ultimate stress levels.

The peripherally distributed configuration exhibits superior stress redistribution capacity compared to centrally concentrated arrangements [29]. The enlarged interfacial contact area between outer carbon fiber layers and inner glass fibers provides multiple load transfer paths, preventing localized stress concentrations that could cause premature failure.

During segment *cd*, the stress maintains the ultimate strength of HFRP bars, expressed as:

$$\sigma_c = \sigma_y^H \tag{16}$$

where σ_y^H represents the ultimate strength of HFRP bars.

Failure occurs in HFRP bars when the ultimate strain of glass fibers is attained, causing deactivation of remaining load-bearing glass fibers.

5.2. Summary of equations

Based on the five-phase constitutive model established by Gao et al. [28], a modified tensile constitutive model differentiating fiber distribution patterns is proposed, incorporating visualized simulation results. The derived governing equations are formulated as follows:

Carbon fiber outsourcing distribution:

$$\sigma = \begin{cases} \left[V_f(V_LE_L + V_HE_H) + E_M\left(1 - V_f\right)\right]\varepsilon, 0 \le \varepsilon \le \varepsilon_a \\ \left[\sigma_y, \quad \sigma_b\right], \varepsilon = \varepsilon_a = \varepsilon_b, \\ \alpha_1 V_f(\beta_1 V_LE_L + V_HE_H)\varepsilon, \varepsilon_b \le \varepsilon \le \varepsilon_c \\ \sigma_y^H, \varepsilon_c \le \varepsilon \le \varepsilon_d \end{cases} \tag{17}$$

Concentrated distribution of carbon fibers:

$$\sigma' = \begin{cases} \left[V_f(V_L E_L + V_H E_H) + E_M(1 - V_f) \right]\varepsilon, 0 \le \varepsilon \le \varepsilon_{a'} \\ \left[\sigma_y, \quad \sigma_b \right], \varepsilon = \varepsilon_{a'} = \varepsilon_{b'} \\ \alpha_2 V_f(\beta_2 V_L E_L + V_H E_H)\varepsilon, \varepsilon_{b'} \le \varepsilon \le \varepsilon_{c'} \end{cases} \tag{18}$$

In previous studies, the stress recovery phase was generally analyzed using experimental data, where empirical formulas for SSDR (Sudden Stress Drop Rate of HFRP bars during post-yield phase) were typically derived through regression. This study innovatively addresses theoretical deficiencies in regression models by implementing visual tracking of fiber unit failures through numerical simulations, coupled with a modified coefficient for E_{Hf} derived from hybrid constitutive laws.

Fig. 11 compares the $bc/\ b'c'$ segments between the proposed constitutive model and the five-phase model. The slope of each curve represents the elastic modulus during stress recovery. The proposed model, accounting for fiber distribution patterns, demonstrates superior alignment with experimental results compared to the conventional five-phase model. Experimental data show that H_1FRP bars exhibit an average E_{Hf} of 27.69 GPa, while simulation-predicted values reach 27.83 GPa—an 11.5 % accuracy improvement over the five-phase model prediction (31.02 GPa). For H_2FRP bars, experimental E_{Hf} averages 38.18 GPa versus simulated 38.8 GPa, achieving 17.1 % higher accuracy than the five-phase model (31.02 GPa).

These results confirm the significant influence of fiber distribution patterns on HFRP tensile behavior. Incorporating shear-lag equations to address transverse shear effects further enhances prediction accuracy, as validated by the visualized simulation outcomes.

6. Conclusions

This study employed mechanical tests, microstructural observations, and numerical simulation analysis to investigate the effects of different types of fiber distribution variations on the tensile performance of HFRP bars. The investigation revealed that distinct fiber distribution patterns significantly influenced the tensile strength and ductility of HFRP bars. Numerical simulation results demonstrated strong agreement with experimental data, while the load transfer mechanisms among fibers were elucidated. The main conclusions are summarized as follows.

1. The tensile performance of HFRP bars is significantly influenced by fiber distribution patterns. H_1FRP bars with peripheral carbon fiber distribution exhibited 20 % lower tensile strength compared to centrally concentrated H_2FRP bars. Notably, their stress-strain curves demonstrated oscillatory characteristics in the ultimate strength region, revealing a pseudo-yield phenomenon that effectively mitigated material brittleness.
2. The internal damage severity of HFRP bars decreases gradually along the radial direction (from outer to inner layers). The distribution positions of fiber layers and interface of different types of fibers, were found to significantly influence their tensile performance. The average amplitude values of three microstructure damages in H_1FRP bars decreased by approximately 90 % relative to H_2FRP and GFRP counterparts. Furthermore, X-ray CT quantification showed significantly lower residual volume ratios in peripheral fibers compared to internal fibers. Post-failure fiber bundle measurements revealed greater carbon/glass fiber separation in H_2FRP than in H_1FRP. SEM characterization further revealed pronounced weak interfacial bonding at carbon/glass fiber junctions.
3. The load transfer efficiency between different fibers is identified as the critical influencing mechanism. Load transfer is primarily facilitated through transverse shear stress. The finite element analysis indicated a 12.4 % reduction in ultimate tensile strength for H_1FRP relative to H_2FRP. Post-carbon-fiber failure, only 60 % of glass fibers exhibited stress escalation. Transverse shear forces disrupted inter-fiber stress transfer, reducing utilization efficiency of high-elongation fibers. In H_2FRP, 28.8 % of glass fibers sustained damage during carbon fiber failure due to absence of peripheral confinement, leading to brittle fracture from insufficient residual load-bearing capacity.
4. The proposed tensile constitutive model incorporating fiber distribution patterns improved stress recovery phase prediction accuracy by 15 %, demonstrating the critical role of fiber architecture and shear modeling in HFRP tensile performance.

The conceptual framework of fiber transition interfaces and monomorphic fiber layers, when integrated with bio-inspired gradient structural design, may offer novel research pathways for developing hierarchically alternating HFRP composites with enhanced tensile performance.

CRediT authorship contribution statement

Zuquan Jin: Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Mingfei Xu:** Writing – original draft, Visualization, Validation, Data curation. **Bo Pang:** Supervision, Methodology, Funding acquisition. **Yubin Cao:** Supervision, Formal analysis. **Hong Wang:** Validation, Investigation.

Declaration of competing interest

The authors declare no conflict of interest on this manuscript entitled with "**Role of fiber distribution on tensile properties of**

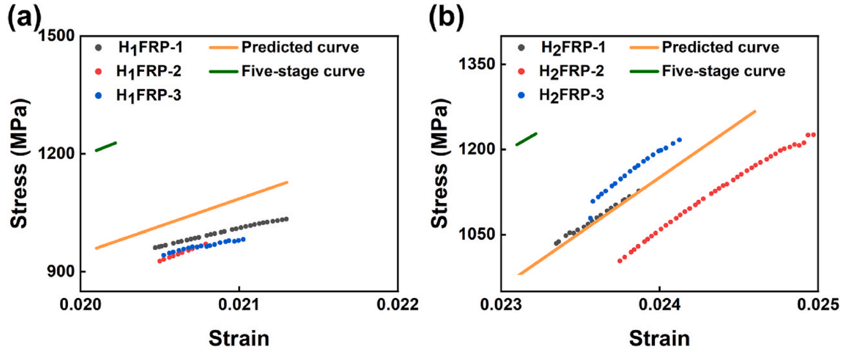

Fig. 11. Prediction of *bc*/ *b′c′* segments in HFRP bar tensile constitutive modeling of (a) H_1FRP bars; (b) H_2FRP bars.

hybrid fiber-reinforced polymer (HFRP) bars through experimental and numerical simulation study".

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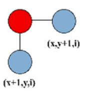
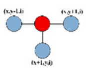
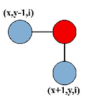
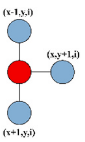
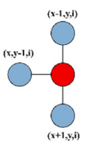
Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jobe.2025.114750.

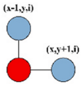
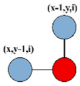
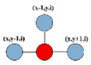
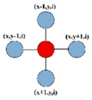
Appendix

Appendix Table 1
Shear-lag equation parameters of fiber joints at different positions.

Fiber position	A	B	C	D	E	F
	$a\varphi_{x,y,i}$	$a(\varphi_{x,y,i}+\varphi_{x,y,i-1})+b\frac{G_m}{d}(\varphi_{x,y+1,i-1}+\varphi_{x+1,y,i-1})$	$a\varphi_{x,y,i-1}$	$b\tau_r(2-\varphi_{x,y+1,i-1}-\varphi_{x+1,y,i-1})$	$b\frac{G_m}{d}$	$\varphi_{x,y+1,i-1}u_{x,y+1,i-1}+\varphi_{x+1,y,i-1}u_{x+1,y,i-1}$
	$a\varphi_{x,y,i}$	$a(\varphi_{x,y,i}+\varphi_{x,y,i-1})+b\frac{G_m}{d}(\varphi_{x,y+1,i-1}+\varphi_{x+1,y,i-1}+\varphi_{x,y-1,i-1})$	$a\varphi_{x,y,i-1}$	$b\tau_r(3-\varphi_{x,y+1,i-1}-\varphi_{x+1,y,i-1}-\varphi_{x,y-1,i-1})$	$b\frac{G_m}{d}$	$\varphi_{x,y+1,i-1}u_{x,y+1,i-1}+\varphi_{x+1,y,i-1}u_{x+1,y,i-1}+\varphi_{x,y-1,i-1}u_{x,y-1,i-1}$
	$a\varphi_{x,y,i}$	$a(\varphi_{x,y,i}+\varphi_{x,y,i-1})+b\frac{G_m}{d}(\varphi_{x,y-1,i-1}+\varphi_{x+1,y,i-1})$	$a\varphi_{x,y,i-1}$	$b\tau_r(2-\varphi_{x,y-1,i-1}-\varphi_{x+1,y,i-1})$	$b\frac{G_m}{d}$	$\varphi_{x,y-1,i-1}u_{x,y-1,i-1}+\varphi_{x+1,y,i-1}u_{x+1,y,i-1}$
	$a\varphi_{x,y,i}$	$a(\varphi_{x,y,i}+\varphi_{x,y,i-1})+b\frac{G_m}{d}(\varphi_{x,y+1,i-1}+\varphi_{x+1,y,i-1}+\varphi_{x-1,y,i-1})$	$a\varphi_{x,y,i-1}$	$b\tau_r(3-\varphi_{x,y+1,i-1}-\varphi_{x+1,y,i-1}-\varphi_{x-1,y,i-1})$	$b\frac{G_m}{d}$	$\varphi_{x,y+1,i-1}u_{x,y+1,i-1}+\varphi_{x+1,y,i-1}u_{x+1,y,i-1}+\varphi_{x-1,y,i-1}u_{x-1,y,i-1}$
	$a\varphi_{x,y,i}$	$a(\varphi_{x,y,i}+\varphi_{x,y,i-1})+b\frac{G_m}{d}(\varphi_{x,y-1,i-1}+\varphi_{x+1,y,i-1}+\varphi_{x-1,y,i-1})$	$a\varphi_{x,y,i-1}$	$b\tau_r(3-\varphi_{x+1,y,i-1}-\varphi_{x,y-1,i-1}-\varphi_{x-1,y,i-1})$	$b\frac{G_m}{d}$	$\varphi_{x,y-1,i-1}u_{x,y-1,i-1}+\varphi_{x+1,y,i-1}u_{x+1,y,i-1}+\varphi_{x-1,y,i-1}u_{x-1,y,i-1}$

(*continued on next page*)

Appendix Table 1 (*continued*)

Fiber position	A	B	C	D	E	F
	$a\varphi_{x,y,i}$	$a(\varphi_{x,y,i}+\varphi_{x,y,i-1})+b\frac{G_m}{d}(\varphi_{x,y+1,i-1}+\varphi_{x-1,y,i-1})$	$a\varphi_{x,y,i-1}$	$b\tau_r(2-\varphi_{x,y+1,i-1}-\varphi_{x-1,y,i-1})$	$b\frac{G_m}{d}$	$\varphi_{x,y+1,i-1}u_{x,y+1,i-1}+\varphi_{x-1,y,i-1}u_{x-1,y,i-1}$
	$a\varphi_{x,y,i}$	$a(\varphi_{x,y,i}+\varphi_{x,y,i-1})+b\frac{G_m}{d}(\varphi_{x,y-1,i-1}+\varphi_{x-1,y,i-1})$	$a\varphi_{x,y,i-1}$	$b\tau_r(2-\varphi_{x,y-1,i-1}-\varphi_{x-1,y,i-1})$	$b\frac{G_m}{d}$	$\varphi_{x,y-1,i-1}u_{x,y-1,i-1}+\varphi_{x-1,y,i-1}u_{x-1,y,i-1}$
	$a\varphi_{x,y,i}$	$a(\varphi_{x,y,i}+\varphi_{x,y,i-1})+b\frac{G_m}{d}(\varphi_{x,y+1,i-1}+\varphi_{x-1,y,i-1}+\varphi_{x,y-1,i-1})$	$a\varphi_{x,y,i-1}$	$b\tau_r(3-\varphi_{x,y-1,i-1}-\varphi_{x,y+1,i-1}-\varphi_{x-1,y,i-1})$	$b\frac{G_m}{d}$	$\varphi_{x,y+1,i-1}u_{x,y+1,i-1}+\varphi_{x-1,y,i-1}u_{x-1,y,i-1}+\varphi_{x,y-1,i-1}u_{x,y-1,i-1}$
	$a\varphi_{x,y,i}$	$a(\varphi_{x,y,i}+\varphi_{x,y,i-1})+b\frac{G_m}{d}(\varphi_{x,y+1,i-1}+\varphi_{x+1,y,i-1}+\varphi_{x,y-1,i-1}+\varphi_{x-1,y,i-1})$	$a\varphi_{x,y,i-1}$	$b\tau_r(4-\varphi_{x,y-1,i-1}-\varphi_{x,y+1,i-1}-\varphi_{x+1,y,i-1}-\varphi_{x-1,y,i-1})$	$b\frac{G_m}{d}$	$\varphi_{x,y+1,i-1}\varphi_{x,y+1,i-1}+\varphi_{x+1,y,i-1}\varphi_{x+1,y,i-1}+\varphi_{x-1,y,i-1}\varphi_{x-1,y,i-1}+\varphi_{x,y-1,i-1}\varphi_{x,y-1,i-1}$

$\varphi_{x,y,i}$ is the damage coefficient, with an initial value of 1, which becomes 0 when the corresponding fiber unit is damaged.

Data availability

Data will be made available on request.

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